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Accuracy and processing speed trade-offs in classical and quantum SVM classifier exploiting PRISMA hyperspectral imagery

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ABSTRACT

Quantum machine learning (QML) focuses on machine learning models developed explicitly for quantum computers. Availability of the first quantum processor led to further research, particularly exploring possible practical applications of QML algorithms in the remote sensing field. The demand for extensive field data for remote sensing applications has started creating bottlenecks for classical machine learning algorithms. QML is becoming a potential solution to tackle big data problems as it can learn from fewer data. This paper presents a QML model based on a quantum support vector machine (QSVM) to classify Holm Oak trees using PRISMA hyperspectral Imagery. Implementation of quantum models was carried on a quantum simulator and a real-time superconducting quantum processor of IBM. The performance of the QML model is validated in terms of dataset size, overall accuracy, number of qubits, training and predicting speed. Results were indicative that (i) QSVM offered 5% higher accuracy than classical SVM (CSVM) with 50 samples and ≥12 qubits/feature dimensions whereas with 20 samples at 16 Qubits/feature dimension, (ii) training time for QSVM at maximum accuracy was 284 s with 50 samples and with 20 samples was 53.68 s and (iii) predicting time for 400 pixels using the QSVM model trained with 50 samples dataset was 5243 s whereas with 20 samples dataset was 2845 s. Results were indicative that QML offers better accuracy but lack training and predicting speed for hyperspectral data. Another observation is that predicting speed of QSVM depends on the number of samples used to train the model.

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Classification; hyperspectral imagery; IBM simulator; PRISMA; quantum; support vector machine

1. Introduction

Quantum machine learning (QML) is an interdisciplinary field of Quantum Mechanics, and Machine Learning outperforms classical computation by outsourcing complex computations to a quantum computer (Saini et al. 2020). Still, researchers worldwide are trying to implement QML for different applications, specifically for big data applications, to evaluate the performance compared to classical computation. At present, two major providers of cloud QC environments are D-Wave (Cavallaro 2021; Otgonbaatar and Datcu 2021a)

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and IBM systems (Saini et al. 2020; Aaron et al. 2021). At present, searching for potential applications that demonstrate quantum speed-ups is an important goal of this field. Specifically, the quantum advantage in machine learning is being tested due to its broad applicability (Arunachalam and de Wolf 2017; Biamonte et al. 2017a; Ciliberto et al. 2018). Considering this, several algorithms have been proposed that promise exponential and polynomial speed-ups with an assumption that classical data are encoded in the amplitudes of a quantum state, and this approach depends upon the number of qubits that is only logarithmic in the size of the dataset. Therefore, quantum algorithms can achieve exponential speed-ups over classical approaches (Harrow, Hassidim and Lloyd 2009; Wiebe, Braun and Lloyd 2012; Lloyd, Mohseni and Rebentrost 2013; Rebentrost, Mohseni and Lloyd 2013).

However, it is unknown whether data can be efficiently provided in practically relevant settings, and this raises the question of whether the advantage comes from the quantum algorithms or the way data are provided (Aaronson 2015). If classical algorithms have equal sampling access to data, then some of the proposed exponential speed-ups no longer exist (Biamonte et al. 2017b; Arunachalam and de Wolf, 2017.; Ciliberto et al. 2018; Lloyd, Mohseni and Rebentrost 2014; Cong and Duan 2015; Zhao, Fitzsimons and Fitzsimons 2015; Farhi and Neven 2018; Chia et al. 2020; Tang 2018; Zhao et al. 2021). In the last decade, quantum computing has attracted many researchers worldwide due to the development of prototypes of quantum computers (Otgonbaatar and Datcu 2021b).

In this comparative study, for the purpose of demonstration, Holm Oak trees were chosen for the classification analysis. From the literature (Caudullo et al. 2016), it was understood that these trees are one of the significant wildfire fuel types available in Sardinian Monte-Arcosu Forest. Due to the declining trend of Holm Oaks, long-term monitoring using phenological Landsat imagery for 34 years of time series data were developed by Valentine, Amaral Paulo and Silva (2019). However, the hyperspectral remote sensing data are often preferred for species level classification due to hundreds of bands at narrow bandwidths. As a part of the S2IGI, An Integrated System for Wildfire Prevention and Management project, Shaik, Laneve and Fusilli (2022) have developed a semi-supervised approach for classifying different fuel types. Vangi et al. (2021) have conducted a study to classify forest types available in the Tuscany region of Italy, among which Oak trees were also classified but not using machine learning. For hyperspectral image analysis, specifically for vegetation classification (Spiller et al. 2021), machine learning classifiers are the primary option for most remote sensing researchers due to their incredible predictive power (Gewali, Monteiro and Sabre 2018).

It is known that the accuracy of any machine learning techniques depends upon the magnitude of the training dataset. For classical machine learning, large datasets are required to obtain higher accuracy, so annotated datasets are a crucial pre-condition for modelling and validating machine learning-based classifications. Unfortunately, due the heterogeneity of remote sensing measurements and tasks, there is not a single go-to-dataset, which could serve the purpose of standardized pre-training and benchmark (Schmitt, Ali Ahmadi and Hänsch 2021), but there are procedures to develop datasets on their own in a semi-supervised approach (Uddien Shaik, Giovanni and Fusilli 2021). A small dataset can be prepared with higher confidentiality using such semi-supervised approaches.

Support vector machine (SVM) is a popular and widely used machine learning model for supervised classification. It has a unique feature of getting trained with relatively a smaller number of samples which attracts most users (Tang 2019, 2018; Childs 2009). Specifically, in remote sensing, it is challenging to obtain ground data as it is expensive to conduct field observations across different locations of the study region to ensure the high spatial variability in the sampling points, particularly at the same of satellite data acquisition. Its effectiveness has been proven in several applications and with different kinds of remote sensing data such as multispectral (García et al. 2011), microwave (Xiao et al. 2018), hyperspectral (Gewali, Monteiro and Sabre 2018; Archibald and Fann 2007; Christovam et al. 2019; Guo et al. 2019; Du, Tan and Xing 2012), etc. SVM algorithm generally is based upon the idea of finding an optimum hyperplane using which the given data are possibly divided into two or more classes. Accuracy decreases when it becomes challenging to find the optimum hyperplane; in such cases, it is necessary to move from two-dimensional to three-dimensional space and use the appropriate kernel functions (Rebentrost, Mohseni and Lloyd 2013, 2014). When the data pattern is complex, it requires more dimensional planes to solve the problem, which is hard for classical computers. The complex data look like random noise for classical computers, whereas quantum computers can recognize it as a labelled pattern (Liu, Arunachalam and Temme 2021).

From various literature studies, it was observed that the quantum SVM could effectively learn or train the system with a smaller dataset than the classical SVM (Huang et al. 2021). The SVM with the quantum kernel can solve the problem considering higher dimensional planes or feature spaces. Previously, few researchers have made a comparative study on classical and QML, where most of them have focused only on the accuracy of the model/classification. For example, Cavallaro et al. (2021) have conducted a comparative study on multispectral imagery and obtained higher accuracy for quantum support vector machine (QSVM) on D-wave quantum annealers. Their study concluded that QML is the alternative when limited data are available. Saini et al. (2020) compared the different backends provided by IBM quantum (IBMQ) in terms of accuracy and execution time using the breast cancer dataset available in the scikit-learn library and concluded that classical machine learning models give better results, but quantum computers are faster. Otgonbaatar and Datcu (2021a) in their study used quantum annealer for subset feature selection and the classification of Indian pines hyperspectral data were unable to obtain better results than the conventional methods. They concluded that the choice and size of the dataset play an essential role in quantum computers and will focus on that in their future study. Specifically, there is no literature reported on quantum accuracy and speed on hyperspectral data.

Quantum annealing (QA) and gate-based are two different paradigms, and their implementation is also different (Otgonbaatar and Datcu 2021b, 2021c). Machine learning on D-wave quantum systems works by quantum annealing algorithm, which utilizes the quantum tunnelling effects based on the adiabatic computing theorem as a natural evolution towards low-energy states. This potential to approximate or even achieve the global optimum is the core advantage of QA compared to the classical methods, which can get trapped in the local optimum in large-scale cases. Quantum

annealing can be a self-organizing and adaptive evolutionary algorithm suitable for solving searching and combinational optimization problems. The intrinsic evolutionary property of QA towards the low-energy states is expected to show strong robustness and versatility in machine learning problems in the aspect of intelligent computing. QML based on a gate-based system is provided by IBMQ and is used in this work. A detailed description of the procedure for implementing this system is given in this manuscript.

Recent studies have demonstrated the efficiency of innovative computing systems like QML in classifying multispectral remote sensing datasets compared with their classical counterparts (Delilbasic et al. 2021; Riedel, Cavallaro and Benediktsson 2021; Zaidenberg et al. 2021; Sebastianelli et al. 2022). However, there is no study on using QSVM for hyperspectral image classification to investigate the accuracy, training and predicting speed for increment in a number of qubits. The present study addressed this major research gap by employing QSVM for PRISMA hyperspectral imagery on IBMQ. The main objectives of this study are first to classify the Holm Oak trees using quantum and classical SVMs with smaller (two different sizes) and similar datasets to evaluate its learning capability for PRISMA hyperspectral data and second to perform a comparative study to understand the trade-off between quantum and classical SVM focusing on the specific parameters, namely, the impact of training samples size, the number of qubits on model accuracy, the training speed of model, and predicting speed. IBM systems' freely available QC environment was considered for implementing QSVM to classify Holm Oak Trees on Sardinian Monte-Arcosu Forest.

This manuscript is organized into four sections – Section 2 describes the study area and PRISMA satellite data implemented in this analysis, Section 3 is dedicated to the methods, Section 4 explains the experimental setup, Section 5 describes the results and discussions in which the trade-offs between classical SVM and quantum SVM were presented, and finally Section 6 explains the conclusions derived from this study.

2. Study area and data

2.1. Study area

For the experimental purpose, hyperspectral data of PRISMA (PRecursore IperSpettrale della Missione Applicativa) on the southwest of Sardinia covering a Monte-Arcosu Forest were considered, as shown in Figure 1. Sardinia is one of the largest islands in the Mediterranean Sea, with frequent fires. Joint Research Commission has reported that, in the last decade, 1005 fires occurred, of which 20% of the total fires occurred national level. The temperature in this area varied from 10°C to 24°C from winter to summer, respectively. Monte-Arcosu Forest is located on a hilly terrain and is one of the largest Holm Oak (*Quercion Ilicis*) trees containing forest. This forest covers an area of 32 km^2 and altitude goes up to 850 m. The highest areas have sparse vegetation, such as meadows of different families, whereas lower areas have a broadleaved forest of *Quercetalia Ilicis* family (Mossa et al. 2016; Caudullo et al. 2016).

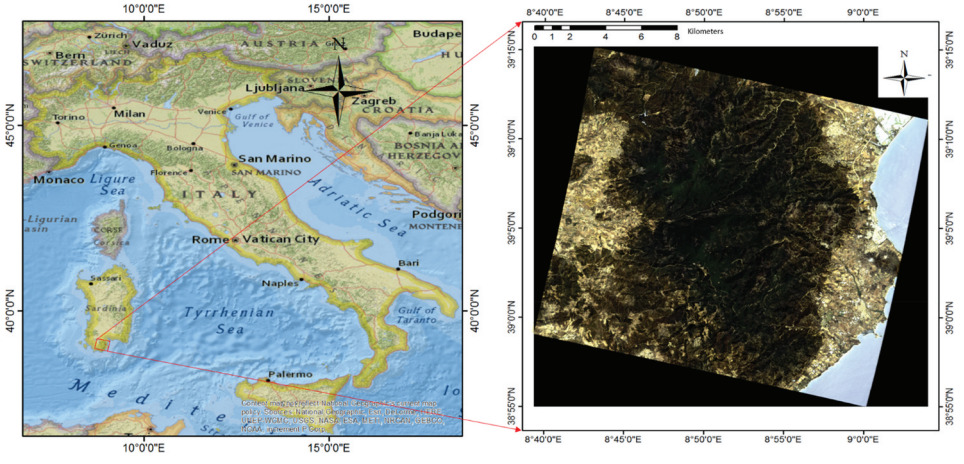


Figure 1. Region of interest – south west of Sardinia covering a part of Monte-Arcosu Forest.

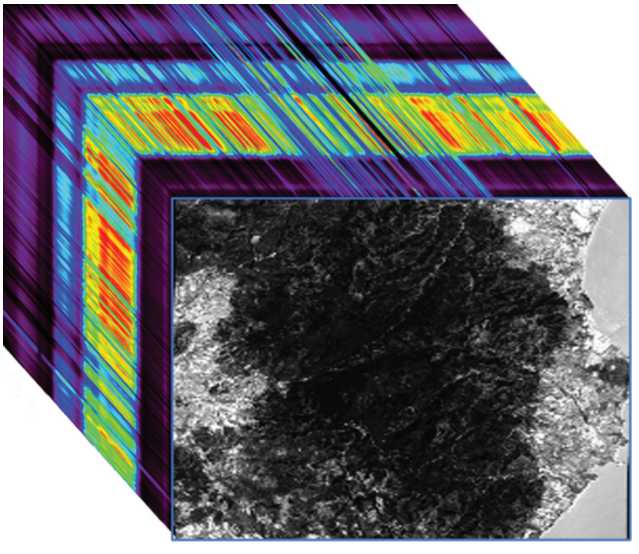


Figure 2. PRISMA hyperspectral cube with 233 bands (VNIR + SWIR).

2.2. PRISMA hyperspectral data

PRISMA, a satellite of the Italian Space Agency (ASI) carrying a hyperspectral sensor, has taken us into the new era of hyperspectral imaging (Pepe et al. 2020; Shaik, Giovanni and Fusilli 2021). This sensor captures imagery having a continuum of spectral bands with 400–2500 nm at a spatial resolution of 30 m using the imaging spectrometer. There are 173 bands in the shortwave infrared (SWIR) within the 920–2500 nm range, and 66 bands are in the visible near-infrared (VNIR) portion of the spectrum (400–1010 nm) as shown in Figure 2. The widths and spectral sampling intervals are ≤12 nm. A panchromatic camera providing a single band (400–700 nm)

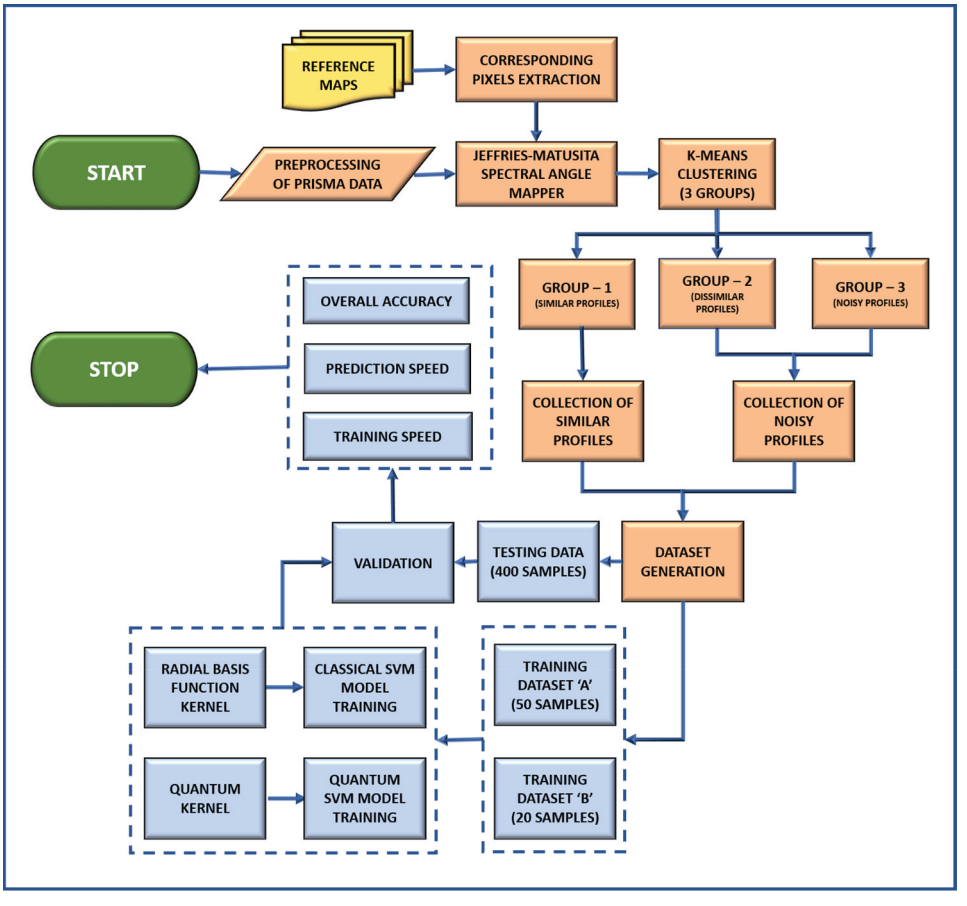


Figure 3. Procedure of samples generation and machine learning.

Table 1. Details of input data.

Dataset name	Sensor	Data points	Train samples	Test samples	Classes
A	PRISMA	600	50	400	2
B	PRISMA	600	20	400	2

image at 5 m spatial resolution is also on-board the ASI's satellite (Amato et al. 2013). The hyperspectral imagery from PRISMA was considered for this comparative study.

PRISMA image has 233 bands of with the size of $1126 \times 1193 \times 233$, and in preprocessing, 37 noisy and water absorption bands were removed by manually selecting. The flowchart shown in Figure 3 explains the procedure of samples generation. First, from the image with $1126 \times 1193 = 13{,}43{,}318$ pixels, a balanced dataset containing 600 pixels was prepared as per the procedure developed by Shaik, Giovanni and Fusilli (2021). Using this, two sub-datasets were formed containing 20 and 50 pixels, respectively, with the help of the scikit-learn package. Then, 400 pixels were taken for validation from the remaining, as shown in Table 1.

Here, pixel represents the feature vector x_n of dimension d = 196, and to each pixel, a label $y_n \in \{0,1\}$ is assigned, indicating the absence or presence of a Holm Oak tree.

3. Methods

3.1. Jeffries Matusita-spectral angle mapper

In hyperspectral remote sensing, a spectral angle measure between reference spectra and target spectra is termed spectral angle, and the technique used for this measurement is spectral angle mapper (SAM) (Padma and Sanjeevi 2014). The smaller the angle, the higher the similarity.

$$\alpha = \cos^{-1}\left(\frac{\sum_{i=1}^{C} t_i r_i}{\sqrt{\sum_{i=1}^{C} t_i^2}\sqrt{\sum_{i=1}^{C} r_i^2}}\right) \tag{1}$$

Given the reference spectra, *r* and the test spectra t of length *C*, the SAM score α is calculated as Eq. (1).

One of the limitations of this technique is that it is insensitive to shade and illumination effects. This limitation has more impact when classifying the vegetation types having similar spectra with some offsets (Sennaraj, Rao Nidamanuri and Bremananth 2013). Hence, SAM is used with the stochastic divergence measures (Du, Chang and County 2004).

In the computation of transformed divergence or divergence, the a priori probability of each spectral class is used as weights. The limitation of this approach is that divergence behaves as a function of the normalized distance between the classes, and there will be a considerable difference from the behaviour of the probability of the correct match. Thus, Jeffries-Matusita (JM) distance measure was combined with SAM to identify similar spectra. JM distance between two spectra is the average distance between the spectra. The exponential factor involved in this method gives an exponentially decreasing weight to increasing separation between the spectra, and this approach overcomes the limitation of transformed divergence (Laliberte, Browning and Rango 2012). JM distance measures band-wise information between spectral vectors (Padma and Sanjeevi 2014). JM distance between the target spectra (S_1) and reference spectra (S_2) can be represented as Eq. (2)

$$JM(S_1, S_2) = 2\left(1 - e^{-B}\right) \tag{2}$$

where Bhattacharya distance (B) measuring the mean (m) and variance (V) of the spectral vectors is defined as in Eq. (3)

$$B(S_1, S_2) = \frac{1}{8}(M_{s1} - M_{s2})^T \left[\frac{V_{s1} + V_{s2}}{2}\right]^{-1} (M_{s1} - M_{s2}) + \frac{1}{2}\ln\left[\frac{|V_{s1} + V_{s2}/2|}{\sqrt{|V_{s1}||V_{s2}|}}\right] \tag{3}$$

To identify similar spectra, JM-SAM (TAN) was used in which the deterministic SAM was combined with stochastic JM distance using tangent function as it projects target spectrum and reference spectrum perpendicularly. This method considers the geometrical aspects

(angle, distance) and band-information between the spectral vectors. The least separable distance between the spectral vectors at each band and the least spectral angle between the vectors are considered the best match (Chang 2004; Du, Chang and County 2004).

3.2. K-means clustering method

An unsupervised clustering (K-means) technique can divide unknown objects into several groups. The members in each group have similar characteristics and properties. K-means is a simple and popular machine learning algorithm used for clustering data where the number of clusters (K) is either known, presumed, or indicated beforehand. However, as per the suggestion of Nguyen et al. (Nguyen et al. 2020; Jones et al. 2020; Huo, Guo and Li 2018), it would be better if the number of clusters K is greater than the number of expected or actual classes, so three clusters were formed since it is a binary classification.

A certain number of K initial centroids of the given input data are determined randomly, and then the data are divided into several groups. Finally, the Euclidean distance (d) between data and centroids is calculated Eq. (4)

$$d(X_t, X_\in) = \sqrt{\sum_{u=1}^{n} \left(X_{ut} - X_{\varepsilon u}^2\right)} \tag{4}$$

where X_t *and* $X_\in$ are the input data and given centroids, respectively; l is the number of the properties; and u is the data property.

By the first-time calculation of d, the result obtained is recorded as $D^{(o)} = \left\{D_1^{(o)}, D_2^{(o)}, \ldots D_K^{(o)}\right\}$, and as the relationship shown in (5), the new centroids are updated

$$X_\in^{(m)} = \frac{1}{h_\in^{(m-1)}} \sum_{X_t = D_{t\in}^{(m-1)}} X_t \tag{5}$$

where m is the number of iterations, $X_\in^{(m)}$ is the new centroid, $h_\in^{(m-1)}$ is the amount of data in the new group based on new centroids.

Iterating the above step and ending the calculation process when $D^{(m)} = D^{(m-1)}$ and $\in^{(m)} = h\in^{(m-1)}$ (Jones et al. 2020). The obtained centroids are the clustering centroids of the data. Three groups were formed having similar spectral signatures, different spectral signatures and very noisy spectral signatures, respectively. The dataset was generated by collecting similar and noisy spectral profile and the same used for training classical and quantum SVM classifiers and validation purpose.

3.3. SVM – radial basis function Kernel

There are numerous supervised learning-based algorithms in the artificial intelligence field which can be applied for classification (Guo et al. 2019; Lu and Weng 2007). In the current framework, SVM (with radial function) was applied because of its applicability in training data set to achieve high accuracy irrespective of the size of the dataset and outstanding generalization capability (Guo et al. 2019). This method works on statistical learning theory and the structural risk minimization principle. The main strategy of this

classifier is to find the optimal separating hyperplane with the maximum margin between the classes from the training samples located at the edge of the class distribution. The formula for the output of a linear SVM is shown in Eq. (6)

$$u = w.x - b \tag{6}$$

x is the input vector, and w is the normal vector to the hyperplane. The separating hyperplane forms at the plane u = 0, having the nearest points lying on the planes at u = ±1. The equation for margin m is shown in Eq. (7)

$$m = \frac{1}{||w||_2} \tag{7}$$

$$k(x_i, x_j) = \exp\left(-\sigma\left|\left|x_i - x_j\right|\right|^2\right), \sigma > 0 \tag{8}$$

Since it is non-linearly separable data, kernels are required for training the model. So, in our case, radial function kernels were implemented. These kernels are a family of kernels where smoothening of distance measure is done by radian function (exponential function), as shown in Eq. (8). Unlike the linear kernel, this kernel can handle the problem when the relation between class labels and attribute is non-linear and can map samples into a higher dimensional space (Cervantes et al. 2020; Ghamisi et al. 2017).

3.4. SVM – quantum kernel

Qubit is the quantum system which is a basic unit of quantum computers and can be represented as a state $|\psi\rangle=\alpha|0\rangle+\beta|1\rangle$, where α and β are the coefficients of the linear combination and $|0\rangle$ and $|1\rangle$ are the two basis states, i.e. their quantum superposition. By definition, the possibility of states to be infinite and subject to $\alpha, \beta \in \mathbb{C}$, $|\alpha|^2 + |\beta|^2 = 1$. The qubit's state cannot be directly measured, and the process of measurement applied to the qubit changes the qubit's state. It can become either $|0\rangle$ or $|1\rangle$ with the probability $|\alpha|^2$ and $|\beta|^2$. This state is also called the post-measurement state (Cavallaro 2021). The ability to solve intractable problems of classical computation using the principle of QC is the cornerstone of quantum supremacy (Havlíček et al. 2019). Quantum supremacy has been proven by experimentally demonstrating high theoretical models (Saini et al. 2020; Li et al. 2015; Rebentrost, Mohseni and Lloyd 2014; Uke, Kumar Soni and Rasool 2020).

The significant component of quantum kernel methods that outperform classical classifiers is the feature map that can map d-dimensional classical data points to quantum state non-linearly and plays a vital role in pattern recognition. It is hard to recognize the intrinsic patterns of data in the original space in learning algorithms, but it becomes accessible by mapping into higher dimensional feature space. The main difference between quantum kernels and classical SVMs are twofold: First, feature maps are more expressive than the classical ones because of the high-dimensional feature maps and, second, quantum kernels produce additive noise in their estimation procedure, whereas classical algorithms compute precisely. Therefore, a demonstration is needed for different datasets to understand whether the quantum kernel is taking advantage of the expressivity of feature maps remaining robust against finite sampling noise (Sain 1996).

3.5. Quantum circuits

As mentioned in the previous section, a feature map can map d-dimensional classical data points to quantum state non-linearly and plays a vital role in pattern recognition. IBM QISKIT Aqua is a library package that provides pre-built functions for generating the different types of feature maps, viz., ZFeatureMap, ZZFeatureMap and PauliFeatureMap. In this study, the second-order Pauli Z evolution circuit (ZZFeatureMap) was selected for the analysis since this is a classical non-linear circuit that allows the qubits to interact with each other. The function 'SecondOrderExpansion' has the arguments such as feature dimension (dimension of the input data and the number of qubits), depth (repetitions of the feature map) and entanglement (connectivity with other qubits). During compiling, QISKIT translates the feature map directly into gates by itself, using which it runs on the quantum computer (Havlíček et al. 2019).

QISKIT also provides the predefined function for QSVM for training the model using the provided feature map, training dataset, and testing dataset. QSVM is also similar to the classical SVM, except the kernels are provided by the quantum distribution, and it will minimize the loss by optimizing the parameters. Finally, in order to run, it is required to provide the simulator (local or IBMQ), shots (number of measurements for each qubit), random seeds (seed simulator and seed transpiler) for better reproducibility. For this analysis, feature dimension has varied, which will be explained in the following section, whereas other arguments were constant like depth = 2, linearly entangled, shots = 1024, seed simulator = 10,598 and seed transpiler = 10,598.

4. Experimental setup

In this work, experiments are proposed using both classical and quantum SVM learning algorithms to classify Holm Oak trees. The datasets for these experiments were prepared as shown in Section 2.2. The classical and QML libraries, Matplotlib, NumPy, Scikit-learn, GDAL, Spectral, etc., were used to program. In addition, a special library package called QISKIT provided by IBM for QML was included.

Classical Support Vector Machine was performed using the Scikit-learn library package considering the radial basis function kernel. It is known that the adjustable parameter plays a significant role in the performance of the kernel and should be carefully tuned. If underestimated, the function will lack regularization, and the decision boundary will be highly sensitive to noise in training data, whereas if overestimated, the exponential will behave almost linearly. Furthermore, training data near the boundary is susceptible to noise, so the complete behaviour of SVM depends on the choice of the width parameter.

So, the Bayesian optimization technique was applied to select the optimum values for C and gamma parameters. For this, the hyperopt package was imported to implement the optimization process giving the considerable range for C and gamma parameters. Hyperopt has two algorithms for hyperparameters search: Random search and Tree of Parzen Estimators (Bayesian). The latter algorithm was implemented in this work, providing the inputs of range 1 to 100 and 1e-3 to 10 for C and gamma parameters, respectively. One thousand evaluations (the number of hyperparameter instances) were performed on each experiment. After this optimization process, the final values of parameters were tabulated in Table 2.

Table 2. Values of hyperparameter optimization for each experiment of two datasets.

	Dataset 'A'		Dataset 'B'	
S. No.	C	Gamma	C	Gamma
1	71.854	3.493	70.915	3.219
2	70.652	4.116	46.710	1.702
3	18.463	0.294	96.208	1.271
4	9.584	2.232	1.852	0.105
5	45.772	1.200	34.620	0.429
6	74.883	2.322	61.219	0.439
7	15.658	2.955	38.780	3.254
8	16.348	2.149	70.661	3.415

QSVM was performed using the QISKIT library package considering the quantum kernel. Currently, two leading cloud QC environments provider are D-Wave Systems and IBM (Cavallaro, 2021). Users can access and program on quantum processors and simulators of the IBM Quantum Experience platform. The computational models were implemented on the backend, including a real-time IBM simulator and a superconducting quantum processor installed on the local. QISKIT has three parts in the package, namely, the provider, the backend and the job. The provider can provide access to different backends such as Aer and IBMQ. Using Aer, access to simulator within QISKIT can be availed to run on the local machine, e.g. statevector_simulator, qasm_simulator, unitary_simulator and clifford_simulator, whereas IBMQ provides access to cloud-based backends (Saini et al. 2020).

The backend signifies either a real quantum processor or a simulator used to run the quantum circuit and generate results. The execution state, i.e. whether the model is running, queued or failed, can be found in the third part of QISKIT, the job (Saini et al. 2020). One of the backends of IBMQ is 'ibmq_qasm_simulator' which has the following features, as shown in Table 3 (Saini et al. 2020). This is the freely available, most extensive usage plan of IBMQ, so it was considered for this study.

Experiments were started from 2 qubits with an increment of 2 qubits till the maximum accuracy of 1 was achieved for quantum accuracy. Then, principal component analysis (PCA) was used to reduce the feature dimensions accordingly to perform this. QISKIT

Table 3. Features of IBM backend.

Properties	ibmq_qasm_simulator
Provider	Ibmq-q/open/main
n_qubits	32
status_msg	Active
backend_version	0.1.547
basic_gates	U1, U2, U3, U, P, R, RX, RY, RZ, ID, X, Y, Z, H, S, SDG, SX, T, TDG, SWAP, CX, CY, CZ, CSX, CP, CU1, CU2, CU3, RXX, RYY, RZZ, RZX, CCX, CSWAP, MCX, MCY, MCZ, MCSX, MCP, MCU1, MCU2, MCU3, MCRX, MCRY, MCRZ, MCR, MCSWAP, UNITARY, DIAGONAL, MULTIPLEXER, INITIALIZE, KRAUS, ROERROR, DELAY
max_circuits	300
max_shots	100,000
max_qubits per pulse gate	3
max_channels per pulse gate	9

Table 4. Specifications of computing resource.

Device specifications	
Computer model	Dell 15 G3 3500
Processor	Intel(R) Core (TM) i7 -10750 H CPU @ 2.60 GHz 2.59 GHz
Installed RAM	16 GB
System type	64-bit operating system, x64-based processor
Number of cores	Six cores/2 computing threads per core
GPU	4 GB Graphics/NVIDIA GEFORCE GTX 1650/120 Hz
Hard drive	1 TB HDD/256 GB SSD
Windows specifications	
Edition	Windows 10 Home Single Language
Version	21H2

Table 5. Trade-offs between QSVM and CSVM for training samples size of 50.

		Overall accuracy		Training time (s)		Predicting Time (Seconds)	
S. No	Qubits	Classical	Quantum	Classical	Quantum	lassical	Quantum
1	2	0.92	0.89	1	39.25	1.2	725.85
2	4	0.92	0.97	1	81.48	1.2	1652
3	6	0.93	0.97	1	158.60	1.2	2644.81
4	8	0.91	0.98	1	183.81	1.2	3478.37
5	10	0.94	0.99	1	293.25	1.2	5207.60
6	12	0.94	1	1	284	1.2	5243
7	14	0.91	1	1	327	1.2	6021
8	16	0.94	1	1	388	1.5	6936

packages on Anaconda and Scikit-learn library in Python were used on the computer with the configuration as shown in Tables 3 and 4. Here, the classical SVM with radial basis function kernel was considered for the comparison.

4.1. The trade-off between classical SVM and quantum SVM for hyperspectral data

Trade-offs between Quantum SVM and Classical SVM were analysed in terms of training samples size {20,50}, accuracy [0 - 1], training time (in seconds) and predicting time (in seconds). Table 5 shows the accuracy and time for predicting and training the 'A' dataset model. Number qubits were increased from 2 to 16 with an increment of 2 till the maximum accuracy has obtained for QSVM. For comparison, classical SVM was also used to obtain accuracy, model training, and predicting times. In classical SVM, qubits represent the feature dimension, which increased similarly to the number of qubits. Predicting time shown in all tabulations is for the test samples size of 400. Similarly, Table 6 shows the accuracies, training and predicting times obtained for the dataset 'B'. Again, to compare with classical and higher samples size, the number of qubits were increased in the same way to check the accuracy.

5. Results and discussions

The proposed experiment employed the QISKIT package on Spyder interface with python 3.8 to write code for the study's models. In this section, results obtained for the experiments were summarized and discussed. Initially, it is to be noted that with an increase in

Table 6. Trade-offs between QSVM and CSVM for training samples size of 20.

		Overall accuracy		Training time (s)		Predicting Time (Seconds)	
S. No	Qubits	Classical	Quantum	Classical	Quantum	Classical	Quantum
1	2	0.89	0.79	1	5	1.2	247
2	4	0.92	0.91	1	10.79	1.2	540
3	6	0.84	0.96	1	17.21	1.2	870
4	8	0.96	0.98	1	23.93	1.2	1287
5	10	0.92	0.97	1	31.67	1.2	1675
6	12	0.94	0.98	1	39.14	1.2	2003
7	14	0.94	0.98	1	46.67	1.2	2306
8	16	0.95	1	1	53.68	1.5	2845

the number of qubits, the accuracy of QSVM also increased (12% at the beginning and 1% while approaching maximum accuracy) while the accuracy increment in classical SVM is not that significant (1 to 2% always).

Figure 3(a and b) depicts the overall accuracy obtained for two different sample sizes (dataset 'A' and 'B') and classical SVM and QSVM. It can be observed from Figure 4a that the highest accuracy of 1 was obtained for QSVM with Qubits equal to 12 for dataset 'A' having 50 samples, whereas for dataset 'B', maximum accuracy obtained at 16 Qubits as seen in Figure 4b. Classical SVM has obtained around 0.95 for both datasets at every feature dimension (Figure 4 a and b). As it is known that quantum technology can solve complex problems, in the same way, it can be observed from this that the quantum kernel can learn the pattern of complex data in a better way than the CSVM with radial basis function kernel.

Here, QSVM and CSVM training speed were compared for different sample sizes and by varying the number of qubits from 2 to 16. Figure 5(a and b) shows the training time for the dataset 'A' and 'B', respectively, for an increase in the number of qubits. For classical SVM, the training time was almost the same (around 1 s) for both datasets. Whereas, for quantum SVM, training time varied from 39.25 to 388 s with an increase in the number of qubits for dataset 'A', and for dataset 'B', training time varied from 5 to 53.68 s shown in Tables 3 and 4. The training speed does not vary much for CSVM but for QSM, the training time increases with number of samples in the dataset.

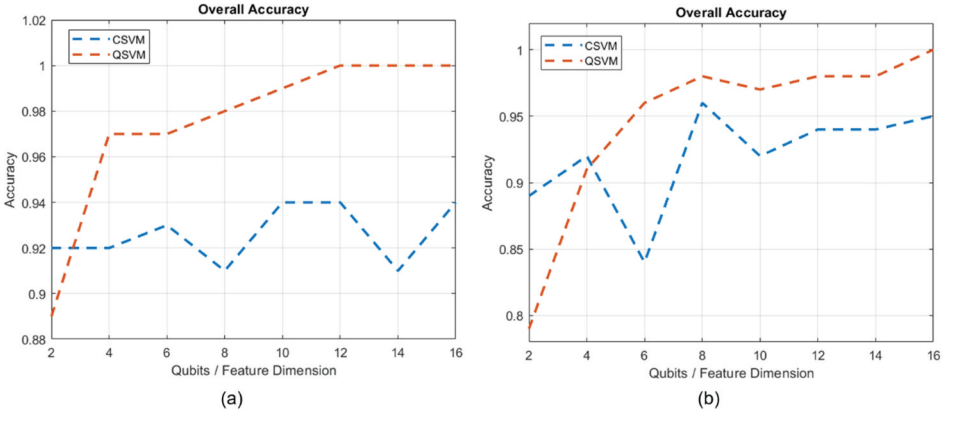


Figure 4. Number of qubits vs overall accuracy: (a) for samples size of 50 and (b) for samples size of 20.

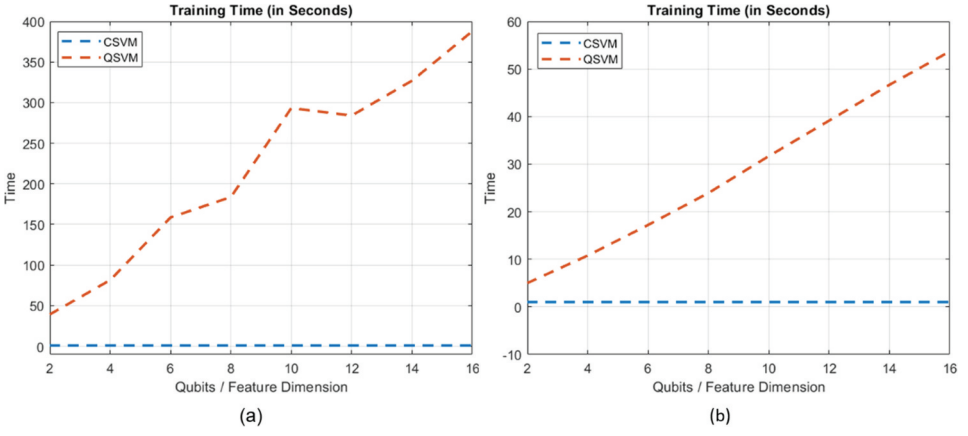


Figure 5. Number of qubits vs training speed: (a) for samples size of 50 and (b) for samples size of 20.

It was observed that predicting speed of QSVM is the main drawback that restricted us from performing the prediction on the whole image. Figure 6(a and b) shows the predicting time obtained on testing data (400 pixels) using the models trained with datasets 'A' and 'B', respectively. The predicting speed remains the same (around 1.5 s) for every possible combination considered for classical SVM. Whereas for the QSVM model, the predicting speed varied from 725.85 to 6936 s, increasing the number of qubits using the model trained with dataset 'A'. Using the model trained with dataset 'B', the predicting time variation was 247 to 2845 s, increasing the number of qubits from 2 to 16, respectively, as shown in Tables 3 and 4.

An increase in feature dimensions (number of qubits/size of samples) increases the superposition value in the order of 2n, where n represents the number of qubits. The developed circuit, which is a feature map in our case, has to be translated into gates to run on a quantum computer. Ness et al. (2021) has commented in their paper that even in the quantum world, gates do not work infinitely fast, and they require a minimum

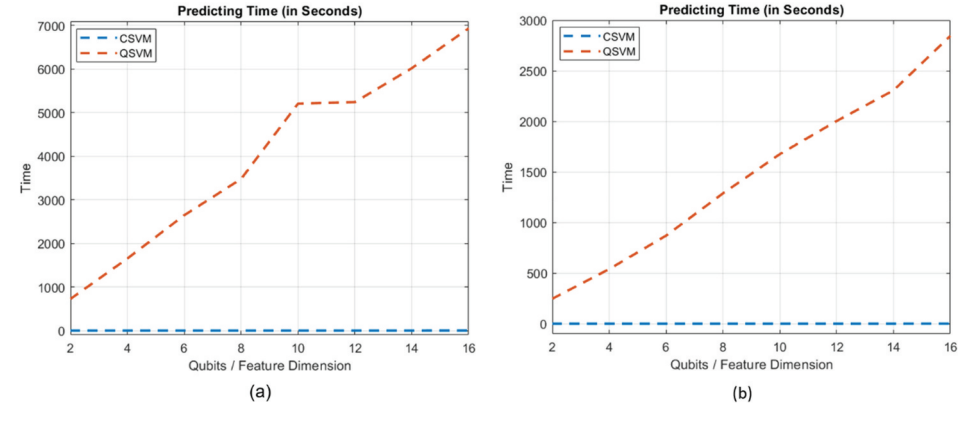


Figure 6. Number of qubits vs predicting speed: (a) for samples size of 50 and (b) for samples size of 20.

amount of time to translate the wave function and the information this contains. Because of this, the processing time has increased with an increase in the number of qubits or gates.

The observed limitation of using QSVM for hyperspectral data is that it cannot be implemented for the whole image containing lakhs of pixels and hundreds of bands because of its higher training and slower executing time. This is in line with other researchers who concluded that quantum computers had solved complex issues with failed data, but exponential speed up is not achieved ultimately (Tang 2019, 2018; Childs 2009). Another observation is that the model trained with higher sample sizes takes more time to predict than the model trained with lesser samples, which does not make sense in classical machine learning.

6. Conclusions

This paper demonstrates the pros and cons of a QML-based QSVM classifier over a CSVM classifier based on radial basis function kernel for applying PRISMA hyperspectral imagery. This comparative study shows that quantum SVM, i.e. quantum kernel, can learn complex data in a better way than the classical SVM with radial basis function kernel. Holm Oak trees were classified with a maximum accuracy of 1 on a quantum computer, whereas the classical SVM on the local machine has 5% lesser accuracy in two different datasets. Results indicate that the accuracy increases with an increase in qubits of a real-time superconducting processor. An observed bottleneck of QML is that the training and predicting speed are incomparable with classical machine learning algorithms. Results anticipate that the QML for hyperspectral data has surpassed solving feature mapping or classification problems in terms of accuracy, and with further developments in quantum technology, exponential speed-up could also be achieved. Though QSVM cannot be used to classify complete imagery having millions of pixels due to its predicting speed limitation, it is soon proposed to be implemented for pseudo-labelling the samples for dataset preparation where there is no single one go-to-dataset available.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability statement

PRISMA imagery that support the findings of this study are accessed from Italian Space Agency's (http://prisma.asi.it/) after the registration (ID: ITA_ScN1_0542_2$).

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